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Intro Quant Methods

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Project Plan and Ethics Review

A constant challenge in urban research is identifying reliable, timely indicators of neighborhood socioeconomic conditions without relying on data sources that are expensive to collect or slow to update. Wang et al. (2017) address this challenge by examining whether 311 service request data can serve as a representative of local urban context, capturing patterns of inequality that traditional measures often miss or report too late. 311 data are non-emergency reports from residents, ranging from graffitis and noise complaints. Wang et al. method involves downloading publicly available 311 complaint records from New York City, Boston, and Chicago (2012–2015), when constructing a "signature vector" for each census tract, a profile of how frequently each complaint type (noise, heating, street conditions, etc.) occurs relative to all requests in that area. They apply k-means clustering to group census tracts by their 311 profiles, then compare those clusters against Census socioeconomic data and Zillow housing prices. Their main findings are that 311-based clusters correspond strongly to socioeconomic differences. Higher-income neighborhoods have distinctly different complaint profiles from poorer ones, and 311 data can even predict future real estate price trends, with Random Forest models achieving R^2 values near 0.80.

PROJECT PLAN

My project will replicate Wang et al.'s approach and apply it to 311 service request data from Detroit, Michigan, a city not included in the original study. My research question driving the project will be, does the structure of 311 service requests reflect neighborhood socioeconomic conditions in Detroit the same way it does in New York, Boston, and Chicago? I will download Detroit's publicly available 311 data (the "Improve Detroit" dataset) from data.detroitmi.gov, and Census socioeconomic data (income, education, unemployment, poverty rates) at the census tract level from the American Community Survey via census.gov. Following Wang et al.'s method, I will construct complaint-type frequency vectors for each census tract, then examine whether neighborhoods with different socioeconomic profiles show systematically different 311 request patterns. Rather than full machine learning regression, I will focus on descriptive comparison of complaint profiles across census tracts grouped by income level, a simpler but still meaningful test of Wang et al.'s core claim.

JUSTIFICATION

While the cities used in Wang et al.'s study, New York, Boston, and Chicago, are large and economically diverse, Detroit carries a different and important historical weight that makes it a uniquely important test case. Beginning in the 1930s, federal redlining practices systematically denied mortgage lending to predominantly Black neighborhoods, concentrating poverty and disinvestment in patterns that affect people to this day. This was heightened by postwar white flight, as middle-class white residents relocated to surrounding suburbs, stripping Detroit of its tax base and leaving behind a city increasingly unable to fund basic services. The combined weight of these forces peaked in Detroit filing for the largest municipal bankruptcy in American history in 2013, followed by ongoing but uneven revitalization that has benefited some

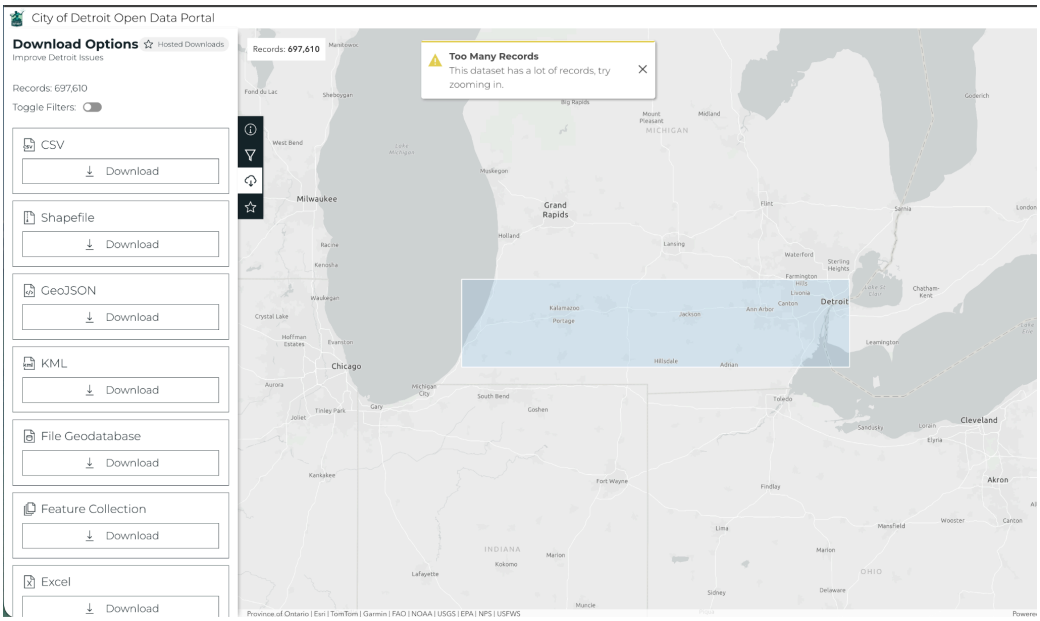
neighborhoods while leaving others further behind. These historical contexts matter for 311 data because, residents in severely disinvested neighborhoods, many of which were deliberately defunded through redlining and decades of municipal neglect, may underreport problems not because conditions are improving, but because generations of unanswered requests have taught them to have distrust in city government. While Wang et al. acknowledges this form of reporting bias, they could not fully test within their sample. However, because of Detroit's historical context, this bias may be more pronounced. Alternatively, 311 patterns might reflect Detroit's revitalization geography, with strong reporting signals in gentrifying areas and weak signals in long-abandoned ones. This study therefore contributes new evidence on whether 311-based urban analytics generalizes beyond stable post-industrial cities, a question with direct implications for the many municipalities considering 311 data as an equity and planning tool.

FEASIBILITY

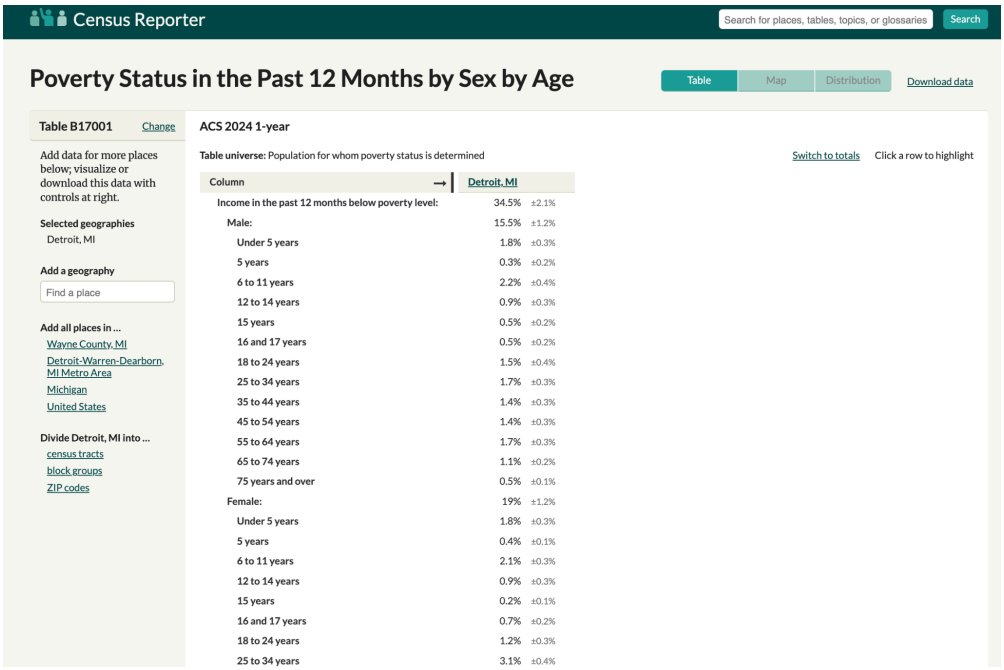
The biggest challenge in this project is that Detroit's 311 system ("Improve Detroit") uses different complaint categories than NYC, Boston, or Chicago, which Wang et al. also encountered across their three cities. To combat this I will be using the closest equivalents used in the original study. Also because I am focusing on the descriptive analysis rather than full cross-country analysis, another challenge would be Detroit having smaller and geographically uneven reporting volume compared to NYC, which may result in some census tracts having too few complaints to analyze fully. I will handle this by grouping the zip code level if needed, as Wang et al. also considered. My backup plan is if the Improve Detroit dataset proves incomplete or poorly formatted, I will use Data Driven Detroit (datadrivendetroit.org), an alternative local data repository that hosts curated Detroit neighborhood datasets. A rough timeline for data

collection and analysis is: download and inspect data by end of this week, clean and organize by end of next week, descriptive analysis in Week 3, write final report in Week 4.

DATA SOURCE



Downloadable options for Detroit 311 data/poverty status



https://data.detroitmi.gov/datasets/9f6987d3fd1b4b7a9d1689ab86ef29e8_0/explore?location=42.

https://censusreporter.org/data/table/?table=B17001&geo_ids=16000US2622000&primary_geo_id=16000US2622000#

ETHICS

This project applies the four ethical principles outlined in Bit by Bit to the use of Detroit 311 data. Respect for Persons is maintained by relying solely on aggregated, publicly available data, avoiding direct human subject involvement. Beneficence is maintained because understanding whether 311 data accurately reflects Detroit's neighborhood could help city planners better target services faster and cheaper. The risks of harm are minimal given the aggregate, public nature of the data. Justice raises concerns that marginalized communities, particularly low-income residents or those with limited trust in government, may be underrepresented, while more civically engaged populations are centered. Respect for Law and Public Interest is upheld through responsible use of open data, there will be no private data, scraping, or API misuse involved. Compared to Urban Informatics and 311 Data, this project takes a more cautious approach by focusing on descriptive analysis rather than predictive modeling. The research primarily benefits planners and researchers but risks reinforcing inequities through misinterpretation or stigmatization of neighborhoods. To mitigate these concerns, I will frame 311 data as incomplete, avoid causal claims, and contextualize findings with Census data. I will not attempt to identify individuals or rank neighborhoods in ways that could cause harm.

AI DISCLOSURE:

This assignment was developed with limited assistance from AI (ChatGPT) for phrasing and organization. All substantive analysis, interpretation, and conclusions are my own.